SurgiGuard AI — Personalized Postoperative Surveillance

Create **SurgiGuard AI**, an AI-assisted platform for **post-discharge postoperative care**. It connects patients, caregivers, and clinical teams to support recovery at home, identify concerning changes, and ensure that alerts receive verified follow-up.

The platform should answer three practical questions: **Is recovery progressing as expected? What needs attention? Has the appropriate follow-up happened?**

1. Personalized recovery plan

Create an individual surveillance plan using the procedure, discharge date, medical history, baseline measurements, relevant risks, and clinician instructions.

Define which measurements, check-ins, photos, and movement tests are needed, when they should occur, and who should receive alerts. Allow clinicians to adjust the schedule as recovery progresses. Distinguish missed measurements from reassuring results.

Include medication context: prescriptions, doses, timing, adherence, reported side effects, and recent changes. Use this context when reviewing symptoms and trends, without automatically changing treatment.

2. Home surveillance

Cover three complementary areas:

- **Respiratory:** IoT oxygen saturation (SpO_2), patient-reported breathlessness, and changes in respiratory symptoms.
- **Hemodynamic:** IoT heart rate, manually entered blood pressure, and symptoms such as dizziness, weakness, or faintness.
- **Neurological and functional:** simple symptom questions and voluntary guided movement tests to follow mobility, coordination, and changes from the patient's baseline.

Collect **IoT temperature readings** alongside these observations. Show measurement timestamps and flag missing, unreliable, or inconsistent readings.

3. Smartphone-guided movement tests

Let patients voluntarily open their **own smartphone camera** and follow short, accessible instructions for clinician-approved movements. Depending on the care plan, examples could

include raising both arms, opening and closing the hands, or performing a seated coordination movement.

Compare observations over time to support review of mobility and neurological recovery. Respect postoperative movement restrictions, allow patients to skip or stop a test, and flag recordings that cannot be interpreted reliably.

No installed camera, passive filming, or continuous video surveillance. Clearly explain camera use and obtain consent before capture or storage.

4. Wound and symptom check-ins

Request **wound photos at dressing changes**, following the patient's existing care instructions. Provide simple guidance for consistent lighting, distance, and framing so clinicians can compare images over time.

Offer **symptom emoji check-ins** for pain, breathing comfort, nausea, fatigue, dizziness, and overall well-being. Pair emojis with clear text labels and optional severity ratings or comments. Ask brief follow-up questions when a response needs clarification.

5. Expected recovery trajectory

Present a simple timeline of expected recovery milestones, personalized to the procedure and clinician-defined plan. Compare actual progress with the patient's baseline and expected course.

Explore predictions of delayed recovery using accumulated observations. Show uncertainty, missing information, and the factors contributing to a prediction. Avoid presenting an estimated recovery date or risk score as a certainty.

6. AI evidence investigations

When a concerning change appears, create a structured investigation that:

- Summarizes what changed, when it began, and whether it is persistent.
- Connects relevant vital-sign trends, symptoms, wound photos, movement observations, and medication context.
- Separates patient reports, device measurements, and AI interpretations.
- Identifies missing or conflicting evidence and requests appropriate clarification within the approved care plan.
- Produces a concise, traceable evidence summary for clinician review.

The investigation should help clinicians assess the situation without presenting an AI-generated explanation as a confirmed diagnosis.

7. Three connected care spaces

Patient mobile app: show today's recovery tasks, measurement instructions, medication reminders, check-ins, scheduled follow-up, and understandable progress. Make it clear whether submitted information is awaiting review.

Caregiver app space: with patient authorization, let caregivers support measurements, reminders, check-ins, and follow-up. Identify entries submitted on the patient's behalf and limit access to the permissions granted.

Clinician web dashboard: provide a patient overview, prioritized alerts, longitudinal measurements, wound-photo comparisons, movement-test observations, medication context, and AI evidence summaries. Allow clinicians to adjust surveillance plans and document decisions.

8. Role-based alerts and verified follow-up

Use clinician-approved rules to route alerts according to urgency and responsibility. Patients receive clear next-step instructions; authorized caregivers receive relevant support requests; clinical teams receive the evidence needed for review.

Track each alert through **creation, assignment, acknowledgement, patient contact, action, and confirmed resolution**. Escalate unacknowledged or overdue alerts according to the agreed workflow. Sending a notification alone must not count as completed follow-up.

9. Prototype scope and boundaries

Demonstrate the complete journey: discharge plan → home observations → recovery comparison → evidence investigation → role-based alert → documented follow-up.

Use simulated or appropriately consented data. Keep the interface accessible, protect sensitive information, and apply role-based access and explicit consent for photos, video, and caregiver participation.

SurgiGuard AI is a prototype for surveillance and clinical decision support, not a diagnostic system. AI interpretations and predictions require clinical validation. The platform must not autonomously prescribe, change medication, or replace clinical assessment or emergency care. Clearly state monitoring availability and response expectations.